# AUSPOS GPS Processing Report

April 14, 2011

This document is a report of the GPS data processing undertaken by the AUSPOS Online GPS Processing Service (version: AUSPOS 2.01) . The AUSPOS Online GPS Processing Service uses International GNSS Service (IGS) products (final, rapid, ultra-rapid depending on availability) to compute precise coordinates in ITRF anywhere on Earth and GDA94 within Australia. The Service is designed to process only dual frequency GPS phase data.

An overview of the GPS processing strategy is included in this report.

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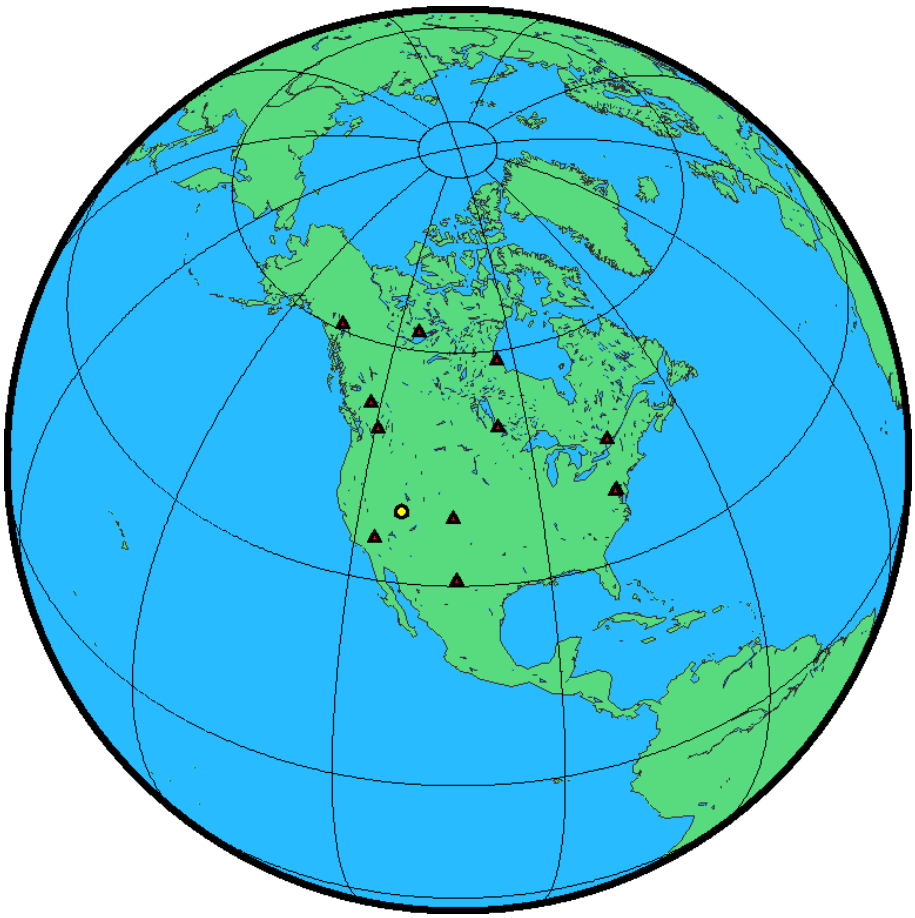
# 1 User Data

All antenna heights refer to the vertical distance from the Ground Mark to the Antenna Reference Point (ARP).

| Station (s) | Submitted File | Antenna Type | Antenna Height (m) | Start Time | End Time |
|---|---|---|---|---|---|
| USER | LR1.11o | TRM41249.00 TZGD | 1.274 | 2011/03/23 15:52:00 | 2011/03/24 00:18:00 |

# 2 Processing Summary



| Date | User Stations | Reference Stations | Orbit Type |
|---|---|---|---|
| 2011/03/23 15:52:00 | USER | AMC2 CHUR DRAO DUBO GODE GOLD MDO1 NRC1 USNO WHIT WILL YELL | IGS final |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# 3 Computed Coordinates, ITRF2005

All computed coordinates are based on the IGS realisation of the ITRF2005 reference frame. All the given ITRF2005 coordinates refer to a mean epoch of the site observation data. All coordinates refer to the Ground Mark.

## 3.1 Cartesian, ITRF2005

| Station | X (m) | Y (m) | Z (m) | ITRF2005 @ |
|---|---|---|---|---|
| USER | -1943693.201 | -4552383.580 | 4011197.049 | 23/03/2011 |
| AMC2 | -1248596.293 | -4819428.212 | 3976505.977 | 23/03/2011 |
| CHUR | -236438.982 | -3307616.888 | 5430049.283 | 23/03/2011 |
| DRAO | -2059164.908 | -3621108.406 | 4814432.302 | 23/03/2011 |
| DUBO | -417603.809 | -4064529.822 | 4881432.165 | 23/03/2011 |
| GODE | 1130773.649 | -4831253.591 | 3994200.460 | 23/03/2011 |
| GOLD | -2353614.431 | -4641385.295 | 3676976.416 | 23/03/2011 |
| MDO1 | -1329998.858 | -5328393.376 | 3236504.134 | 23/03/2011 |
| NRC1 | 1112777.093 | -4341475.873 | 4522955.856 | 23/03/2011 |
| USNO | 1112189.672 | -4842955.042 | 3985352.299 | 23/03/2011 |
| WHIT | -2218338.052 | -2201205.118 | 5543057.514 | 23/03/2011 |
| WILL | -2084258.261 | -3313873.004 | 5019853.031 | 23/03/2011 |
| YELL | -1224452.798 | -2689216.166 | 5633638.287 | 23/03/2011 |

## 3.2 Geodetic, GRS80 Ellipsoid, ITRF2005

Geoid-ellipsoidal separations, in this section, are computed using a spherical harmonic synthesis of the global EGM2008 geoid. More information on the EGM2008 geoid can be found at http://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/

| Station | Latitude (DMS) | Longitude (DMS) | Ellipsoidal Height(m) | Derived Above Geoid Height(m) |
|---|---|---|---|---|
| USER | 39 12 28.32837 | -113 07 14.24518 | 1537.873 | 1557.884 |
| AMC2 | 38 48 11.24698 | -104 31 28.54166 | 1911.402 | 1930.148 |
| CHUR | 58 45 32.67942 | -94 05 19.42757 | -19.277 | 28.888 |
| DRAO | 49 19 21.42622 | -119 37 29.94041 | 541.892 | 558.440 |
| DUBO | 50 15 31.71174 | -95 51 58.25586 | 245.293 | 274.986 |
| GODE | 39 01 18.22035 | -76 49 36.59264 | 14.524 | 47.789 |
| GOLD | 35 25 30.56099 | -116 53 21.30491 | 986.668 | 1017.360 |
| MDO1 | 30 40 49.83921 | -104 00 53.97795 | 2004.502 | 2026.581 |
| NRC1 | 45 27 14.98616 | -75 37 25.78565 | 82.521 | 116.053 |
| USNO | 38 55 08.26701 | -77 03 58.41438 | 48.885 | 81.752 |
| WHIT | 60 45 01.84005 | -135 13 19.60834 | 1427.407 | 1419.592 |
| WILL | 52 14 12.72205 | -122 10 04.12793 | 1095.710 | 1110.454 |
| YELL | 62 28 51.21572 | -114 28 50.53330 | 180.947 | 207.662 |


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# 4 Solution Information

## 4.1 Coordinate Precision - Geodetic, One Sigma

| Station | $\sigma$ East (m) | $\sigma$ North (m) | $\sigma$ Up (m) |
|---|---|---|---|
| USER | 0.002 | 0.002 | 0.003 |
| AMC2 | 0.002 | 0.002 | 0.004 |
| CHUR | 0.002 | 0.002 | 0.003 |
| DRAO | 0.002 | 0.002 | 0.003 |
| DUBO | 0.002 | 0.002 | 0.003 |
| GODE | 0.002 | 0.002 | 0.003 |
| GOLD | 0.002 | 0.002 | 0.003 |
| MDO1 | 0.002 | 0.002 | 0.003 |
| NRC1 | 0.002 | 0.002 | 0.003 |
| USNO | 0.002 | 0.002 | 0.004 |
| WHIT | 0.002 | 0.002 | 0.003 |
| WILL | 0.002 | 0.002 | 0.003 |
| YELL | 0.002 | 0.002 | 0.003 |

## 4.2 Ambiguity Resolution - per baseline

| Baseline | Ambiguities Resolved | Baseline Length (km) |
|---|---|---|
| DRAO - WILL | 88.9 % | 370.4 |
| DRAO - GOLD | 79.2 % | 1556.1 |
| CHUR - YELL | 89.7 % | 1183.2 |
| CHUR - DUBO | 91.3 % | 952.2 |
| GOLD - USER | 76.0 % | 536.3 |
| WHIT - YELL | 79.5 % | 1110.9 |
| GODE - USNO | 95.2 % | 23.7 |
| GOLD - MDO1 | 63.6 % | 1309.1 |
| WILL - YELL | 85.2 % | 1227.3 |
| DUBO - NRC1 | 87.5 % | 1596.0 |
| AMC2 - MDO1 | 79.2 % | 901.8 |
| GODE - NRC1 | 83.3 % | 721.0 |
| AVERAGE | 83.2% | 957.3 |

Please note for a regional solution, such as used by AUSPOS, an average ambiguity resolution of 50% or better for the network indicates a reliable solution.


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# 5 Computation Standards

## 5.1 Computation System

| | |
|---|---|
| Software | Bernese GPS Software Version 5.0. |
| GNSS system(s) | GPS only. |

## 5.2 Data Preprocessing and Measurement Modelling

| | |
|---|---|
| Data preprocessing | Phase preprocessing is undertaken in a baseline by baseline mode using triple-differences. In most cases, cycle slips are fixed by the simultaneous analysis of different linear combinations of L1 and L2. If a cycle slip cannot be fixed reliably, bad data points are removed or new ambiguities are set up A data screening step on the basis of weighted postfit residuals is also performed, and outliers are removed. |
| Basic observable | Carrier phase with an elevation angle cutoff of 10° and a sampling rate of 3 minutes. However, data cleaning is performed a sampling rate of 30 seconds. Elevation dependent weighting is applied according to $1/\sin(e)^2$ where $e$ is the satellite elevation. The code observable is only used for the receiver clock synchronisation. |
| Modelled observable | Double differences of the ionosphere-free linear combination. |
| Ground antenna phase centre calibrations | IGS05 absolute phase-centre variation model is applied. |
| Tropospheric Model | A priori model is the Saastamoinen-based hydrostatic mapped with the dry-Niell. |
| Tropospheric Estimation | Zenith delay corrections are estimated relying on the wet-Niell mapping function in intervals of 2 hour. N-S and E-W horizontal delay parameters are solved for every 24 hours. |
| Tropospheric Mapping Function | Niell |
| Ionosphere | First-order effect eliminated by forming the ionosphere-free linear combination of L1 and L2. |
| Tidal displacements | Solid earth tidal displacements are derived from the complete model from the IERS Conventions 2003, but ocean tide loading is not applied. |
| Atmospheric loading | Not applied |
| Satellite centre of mass correction | IGS05 phase-centre variation model applied |
| Satellite phase centre calibration | IGS05 phase-centre variation model applied |
| Satellite trajectories | Best available IGS products. |
| Earth Orientation | Best available IGS products. |


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## 5.3 Estimation Process

| | |
|---|---|
| Adjustment | Weighted least-squares algorithm. |
| Station coordinates | Coordinate constraints are applied at the Reference sites with standard deviation of 5mm and 10mm for horizontal and vertical components respectively. |
| Troposphere | Zenith delay parameters and pairs of horizontal delay gradient parameters are estimated for each station in intervals of 2 hour and 24 hours. |
| Ionospheric correction | An ionospheric map derived from the contributing reference stations is used to aid ambiguity resolution using the QIF strategy |
| Ambiguity | Ambiguities are resolved in a baseline-by-baseline mode using Quasi-Ionosphere-Free (QIF) approach. |

## 5.4 Reference Frame

| | |
|---|---|
| Terrestrial reference frame | IGS05 station coordinates and velocities mapped to the mean epoch of observation. |
| Australian datum | GDA94 coordinates determined via Helmert transformation from ITRF using the Dawson and Woods (2010) parameters. |
| Derived AHD | For stations within Australia, AUSGeoid09 is used to compute AHD. AUSGeoid09 is the Australia-wide gravimetric quasigeoid model that has been a posteriori fitted to the Australian Height Datum |
| Above-geoid heights | Earth Gravitational Model EGM2008 released by the National Geospatial-Intelligence Agency (NGA) EGM Development Team is used to compute above-geoid heights. This gravitational model is complete to spherical harmonic degree and order 2159, and contains additional coefficients extending to degree 2190 and order 2159. |


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